Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/17/2026
PI Name:	Gangopadhyay, Anup		
Project Title:	NASA Research Under Project QUASI and ESA-Based Investigations THERMOLAB-ISS and ICOPROSOL (PI: Gangopadhyay)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Hanson, Ian	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	Co-Investigator Anup Gangopadhyay, Ph.D. has taken over as Principal Investigator (PI) for this investigation. The previous PI, Kenneth Kelton, has passed away (Ed., 4/9/26).		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K1649		
Performance Goal No.:			
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Task Description:	This grant is a continuation of previous grants NNX10AU19G and NNX16AB52G. Funding is provided for all ground-based studies and flight preparation activities by the QUASI team at Washington University in St. Louis (WUSTL). These support QUASI flight investigations on the International Space Station (ISS) and collaborative flight projects with ESA teams ICOPROSOL (scientific coordinator – Dr. Dirk Holland-Moritz, DLR) and THERMOPROP (scientific coordinator – Prof. Dr. Hans Fecht, Ulm University). The flight experiments on the ISS are in designated batches. Batch 1 and batch 2 experiments have now been successfully completed. Batch 3 has almost been completed and Batches 4 and 5 have been approved. Metallic alloy liquids processed in a containerless environment are investigated in all of the proposed measurements for QUASI, THERMOPROP and ICOPROSOL. The primary goals of the flight and ground-based experiments are: 1. An investigation of the role of stirring on crystal nucleation from a supercooled metallic liquid. 2. Measurements of key parameters needed for a quantitative analysis of the nucleation data obtained in the ground-based and ISS studies (driving free energy, specific heat and viscosity) and for investigations of the connection between structural ordering and thermophysical properties. 3. Measurements of the electrical resistivity of metallic liquids as a function of temperature to investigate a proposed correlation between structural changes in the liquid and dynamical properties, and to confirm the existence of a plateau in the temperature dependence of the resistivity that we have recently discovered at high temperatures. 4. Measurements of the thermal conductivity of metallic liquids at high temperature. 5. Measurements of the crystal growth velocity in the ground-based and ISS studies. 6. Studies of the nucleation step in eutectic solidification. Rationale for HRP Directed Research: Measurements of the maximum supercooling of free-cooling metallic liquids that crystallize to phases of a different composition than the liquid, are made in the quiescent environment of the electrostatic levitation facility (WU-BESL) located in our laboratory at Washington University in St. Louis. Long-range diffusion is required for nucleation in these cases. These ground-based results are compared with undercooling studies made as a function of stirring in the ISS-EML. According to the Coupled Flux Model for nucleation, developed by Kelton, the nucleation rate should increase with the stirring rate, since long-range diffusion will become less important. A key goal of our ongoing research is to investigate this. Measurements of the thermophysical properties, including the specific heat, density, surface tension, viscosity, thermal conductivity and electrical resistivity, will be made using the WU-BESL and the ISS-EML, since these facilities have different capabilities for these studies. The specific heat, density, surface tension and viscosity are needed for a quantitative analysis of the nucleation data. Since our earlier studies have demonstrated that ordering in the liquid couples to the nucleation barrier, the undercooling and thermophysical property measurements will be correlated with x-ray synchrotron diffraction results for the supercooled liquids of interest, obtained previously using WU-BESL. Based on our previous studies, the specific heat and electrical resistivity are also valuable for providing structural evidence for a cooperative cross-over behavior in the shear viscosity at high temperatures. When coupled with the measured electrical resistivity, measurements of the thermal conductivity will allow an investigation of the Widemann-Franz law for high temperature metallic liquids, a case where it is not been studied before. Studies of the effects of stirring during the solidification of a eutectic Zr80Pt20 liquid have shown that nucleation proceeds by a non-Classical Nucleation Theory 2-step process, with primary nucleation of an icosahedral quasicrystal phase followed by the decomposition of this phase to the eutectic phase mixture. This raises the question of the nucleation mechanism for binary eutectic phase in general. Since eutectic solidification is important in many applications, we are investigating this process in other eutectic liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes, cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. The combination of the containerless based methods of WU-BESL and the ISS-EML enables these measurements. In addition to their relevance for processes such as additive manufacturing on earth, where nucleation in highly stirred conditions is important, measurements in two different gravitational fields will be important for designing additive manufacturing in space and on the Moon. Finally, past measurements of the electrical resistivity in high temperature liquids have revealed a plateau in the temperature dependence of the resistivity at temperatures near a crossover temperature for the viscosity. This is not predicted by any existing theory. Our measurements will provide valuable information for developing a theoretical understanding of this phenomenon.
	New Principal Investigator (PI): Anup Gangopadhyay, Ph.D. For previous reporting, see Grant 80NSSC21K1649 under Kenneth Kelton, Ph.D. 2026 Update per the PI: The solidification of a liquid is of fundamental importance in everyday life. For example, freezing of water into ice is of paramount importance for the existence of all living species, while solidification of a metal or alloy liquid into a solid is equally important for all industrial applications. Understanding the basic science of this process is therefore important. Solidification is a two-step process. First, small clusters of atoms form spontaneously in the liquid as it is cooled below its melting temperature. These clusters grow or shrink spontaneously until they exceed a critical size. Only then can it

Task Progress:	grow into a solid. Anything smaller than the critical size shrinks. This process is called nucleation. The main objective of this project is to study this process in metallic alloy liquids. Specifically, to investigate how fluid flow (stirring) affects the nucleation process. Many people are aware that if a clean cup or bottle of water is kept in a freezer, often it remains as a liquid, although the temperature in the freezer may be many degrees below the freezing temperature of water (32°F); in contrast, water under natural environment freezes immediately below 32°F. This shows that a liquid need not freeze immediately below the melting temperature. It may stay in the liquid form for many minutes or hours below the melting temperature (¿$). For high melting temperature alloys, such as steel, it is usually not possible under normal circumstances. However, if a steel ball is floated in a vacuum chamber (gas molecules are very few), it may be kept as a liquid drop at several degrees below (¿$), sometimes even a few hundred degrees below, for many minutes, if not hours. The difference is that the floating liquid is not in contact with a foreign object, such as a container under normal circumstances. The containers usually catalyze nucleation followed by growth of solids. Using electric or electromagnetic fields, it is possible to float objects. These are called electrostatic or electromagnetic levitation techniques (ESL and EML). This research group uses these techniques on Earth under gravity, or on the International Space Station (ISS) under microgravity, to investigate the solidification (freezing) behavior of industrially useful metal alloys at temperatures far below their melting temperatures. The molten liquid drops are allowed to cool as far as they can. At some point, it is not possible to stay as a liquid and it solidifies by releasing a large amount of heat, which abruptly raises its temperature (called recalescence). This temperature, called the nucleation temperature, is measured many times, and such data are fitted to theoretical models. Until this group asked the question whether the nucleation process is influenced by fluid flow, no such study was reported. To investigate this, nucleation has been studied either in a quiescent (no fluid motion) liquid using the ESL in our laboratory or under controlled fluid flow on the ISS. In the past one year, we have completed such experiments on two alloys, Ti45Zr45Ni10 and Cu47Zr47Al6 on Earth, and under microgravity on the ISS. The essential result is that stirring (fluid flow) does not seriously affect the nucleation process in the former, but significantly in the latter. Mechanism responsible for these observations is under investigation. This group also measures many useful thermophysical properties of solids and liquids – such as specific heat, viscosity, surface tension, electrical and thermal conductivity – which are rather difficult to measure for high temperature melts because of easy reaction with container materials. Measurements on levitating droplets are free from contamination and provide clean reliable data. These properties are important for basic understanding of these materials and their technological applications. This has been done over the past one year on these two alloys. It needs to be remembered that, in space, liquids do not boil (convective flow is absent) since density has no role there. Therefore, if one tries to make any useful mechanical part from a liquid or try to weld two pieces together in space, the entire solidification process will be different from Earth. Knowledge acquired from investigations by this group will become very important one day when manufacturing in space under microgravity will be necessary either in long spaceflights to Mars or in colonies on the Moon or Mars, when they are established. As in any experimental study, unforeseen problems crop up. Many such challenges were met last year, such as mechanical problems in the transfer of sample in the Cu coils that generate the electromagnetic field, transfer of video images collected during experiments back to Earth, delay in the installation of new hardware to the facility, to name a few. Theoretical challenges in understanding the experimental results are also cropping up as new data are collected from new experiments. For example, it appears that the thermal conductivity of many liquids may not fit our general understanding of the same property for the solids. As in any investigation, surprising results are always welcome since understanding them often opens up new horizons. Therefore, meeting new challenges are always welcomed by most scientists.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Gangopadhyay AK, Kelton KF. "Temperature dependent specific heat measurements of equilibrium, supercooled liquid, glass, and the corresponding crystal of Cu47Zr47Al6 from terrestrial and microgravity experiments on the international space station." Acta Materialia. 2025 Aug 15;295:121201. http://dx.doi.org/10.1016/j.actamat.2025.121201 , Aug-2025
Articles in Peer-reviewed Journals	Gangopadhyay AK, Becker M, Schneider S, Sheng Y, Kelton KF. "Surface Tension and Viscosity of Zr80Pt20 measured on the International Space Station and its implication on nucleation mechanism." Metall. Mater. Trans. A. 2026 Jan;57(1):259-66. https://doi.org/10.1007/s11661-025-08031-9 , Jan-2026